SOI high-voltage LDMOS with novel triple-layer top silicon based on thin BOX

S.D. Hu, L. Zhang, J. Luo, K.Z. Tan, W.S. Chen, P. Gan, X.C. Zhou and Z. Zhu

A novel SOI high-voltage LDMOS with a triple-layer top silicon (TLTS) is investigated. The top silicon layer of the TLTS LDMOS consists of n^- silicon with a p-top layer, p^- silicon in the middle, and n^+ silicon on the interface. On the condition of high-voltage blocking state, the electric fields of the drift region and BOX are modulated and optimised by the triple-layer top silicon, respectively, which induces a high BV of 624 V for the TLTS LDMOS with a thin buried oxide layer (BOX) of 0.4 μm. Compared with several SOI devices, the proposed TLTS LDMOS has a higher figure-of-merit.

Introduction: Improving the breakdown voltage (BV) for silicon on insulator (SOI) high-voltage devices is always a research hot topic in the area of power microelectronics devices [1], and some work has been carried out recently [2–5]. However, the problem of low BV for SOI devices based on separation by implanted oxygen (SIMOX) is still prominent because of the thin buried oxide layer (BOX). In this Letter, a novel SOI high-voltage LDMOS with a triple-layer top silicon (TLTS SOI) is first proposed, which provides a higher BV based on a thin BOX. The influence of structure parameters on the BV is analysed by MEDICI [6]. The performance of TLTS SOI is compared with several SOI or partial-SOI (PSOI) devices with the same thickness of top silicon.

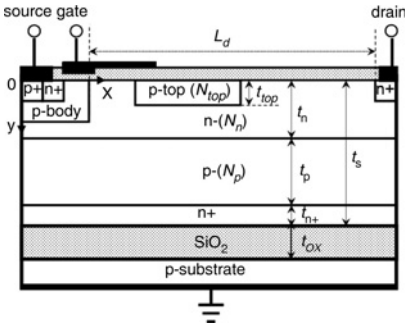


Fig. 1 *Structure of TLTS SOI LDMOS*

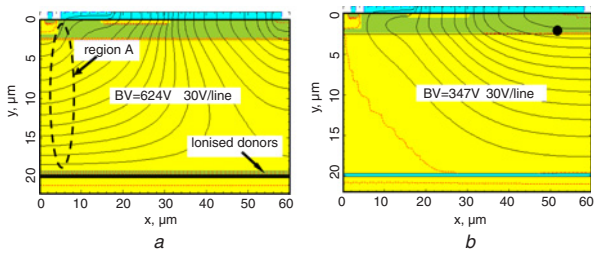


Fig. 2 *Equipotential contours distributions at breakdown ($t_n = 2.5$ μm)*

a Equipotential contours distribution for TLTS SOI LDMOS
$N_n = 3.9 \times 10^{15}$ cm^{-3}, $N_p = 6.5 \times 10^{14}$ cm^{-3}, $N_{top} = 6.1 \times 10^{16}$ cm^{-3}
b Equipotential contours distribution for SOI LDMOS without interface n^+ layer
$N_n = 7.9 \times 10^{15}$ cm^{-3}, $N_p = 6.5 \times 10^{14}$ cm^{-3}, $N_{top} = 1.2 \times 10^{16}$ cm^{-3}

Structure and mechanism: The TLTS SOI LDMOS is shown in Fig. 1. The top silicon layer of the TLTS LDMOS is made of n^- silicon with a p-top layer, p^- silicon in the middle, and n^+ silicon on the interface. Device parameters used in the simulation are listed in Table 1. When a high positive voltage V_d is applied to the drain while the source, gate and substrate are grounded, plenty of ionised donors are formed on the bottom interface of the undepleted n^+ silicon layer, which can effectively enhance the electric field of the BOX by an enhanced dielectric layer field (ENDIF) [7], and at the same time, modulate the electric field of the whole top silicon layer. Adjusting the thickness and the doping concentration of the p^- silicon layer can make the device meet different reduced surface field (RESURF) conditions and get a lower specific on-resistance ($R_{on,sp}$). Figs. 2*a* and *b* show the equipotential contours distributions at breakdown for the TLTS SOI LDMOS with an interface n^+ silicon layer (Fig. 2*a*) and without an interface n^+ silicon layer (Fig. 2*b*). It can be seen that, because of the ionised donors in the n^+ silicon layer, the electric field of the BOX is enhanced, and the equipotential contours distributions of the top silicon are more uniform for the device with the interface n^+ silicon layer than that without an interface n^+ silicon layer. 12 lines are introduced into the region A under the source as shown in Fig. 2*a*, which avoids a premature breakdown at the B-point in Fig. 2*b*. Therefore, a higher BV of 624 V is obtained for the TLTS SOI LDMOS with a thin BOX of 0.4 μm.

Results and discussion: Fig. 3 shows BV against the structure parameters for the TLTS SOI LDMOS. Fig. 3*a* is the dependence of BV and t_n with $N_p = 6.5 \times 10^{14}$ cm^{-3}. It is clear that the maximum BV of 624 V can always be obtained for $t_n = 3.0$, 2.8, 2.5, and 2.3 μm, and different t_n match different N_n by RESURF condition. Fig. 3*b* shows the dependence of BV and N_p with $t_n = 2.5$ μm, and the maximum BV of 624V is achieved when N_p is 6.5×10^{14} cm^{-3}. Considering comprehensively the device performances of BV and $R_{on,sp}$, the optimal structure parameters are $t_n = 2.5$ μm, $N_n = 3.9 \times 10^{15}$ cm^{-3}, and $N_p = 6.5 \times 10^{14}$ cm^{-3}. Table 2 gives the performance comparisons of several SOI or PSOI devices with the same thickness of top silicon layer ($t_S = 20$ μm). TLTS SOI with a thin BOX of 0.4 μm has a higher figure of merit (FOM = BV^2/$R_{on,sp}$). It is especially suitable for use in SIMOX technology because of the thin BOX.

Table 1: Device parameters used in simulation

Parameter	Value
Length of drift region, L_d (μm)	52
Thickness of top silicon, t_S (μm)	20
Thickness of BOX, t_{OX} (μm)	0.4
Thickness of n^- silicon layer, t_n (μm)	Optimised
Thickness of p^- silicon layer, t_n (μm)	Optimised
Thickness of interface n^+ silicon layer, t_{n+} (μm)	0.3
Thickness of p-top layer, t_{top} (μm)	0.5
Concentration of n^- silicon layer, N_n (cm^{-3})	Optimised
Concentration of p^- silicon layer, N_p(cm^{-3})	Optimised
Concentration of interface n^+ layer, N_{n+} (cm^{-3})	1×10^{18}
Doping concentration of p-top layer, N_{top} (cm^{-3})	Optimised

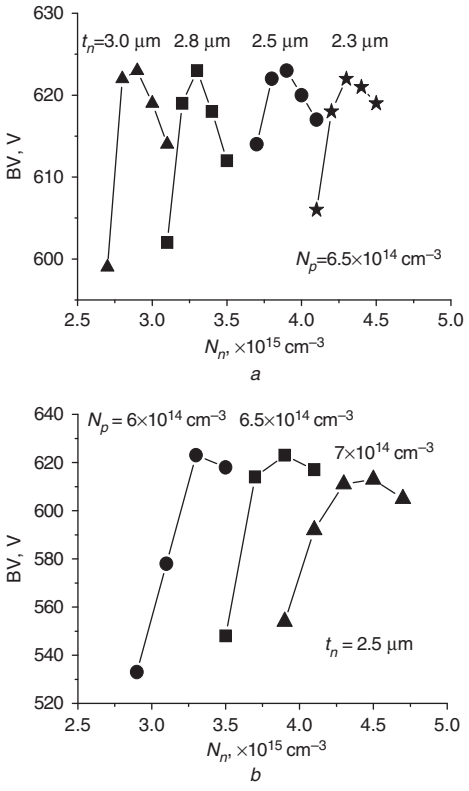


Fig. 3 *BV against structure parameters for TLTS SOI LDMOS*

a BV against t_n ($N_p = 6.5 \times 10^{14}$ cm^{-3})
b BV against N_p ($t_n = 2.5$ μm)

Table 2: Performance comparisons of several SOI or PSOI devices with same t_S of 20 μm

Device		t_{OX}/L_d (μm)	BV (V)	$R_{on,sp}$ (mΩ·cm^2)	FOM (MW·cm^{-2})
SOI	Proposed TLTS SOI	0.4/52	624	206	1.89
	Conventional SOI [3]	3/90	478	575	0.397
	BNL SOI [3]	3/90	548	455	0.66
	DWCBL SOI [4]	3/80	1040	–	–
	CBL SOI [8]	3/80	821	–	–
	Interconnection SOI [9]	5/-	670	–	–
	DT SOI [10]	1/90	796	–	–
PSOI	Conventional PSOI [3]	3/90	548	588	0.511
	BNL PSOI [3]	3/90	660	500	0.871

Conclusion: A novel SOI high-voltage device of the TLTS SOI LDMOS which is especially suitable for use in SIMOX technology is proposed in this Letter. ABV of 624V is obtained based on a thin BOX of 0.4 μm. The TLTS SOI LDMOS has a higher FOM compared with several SOI or PSOI devices.

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